

Context-Aware Video Segmentation & Intelligent Ad Placement

How I thought about it, planned it, built it and made it live, from start to end.

Product: contextaware**adbreaks**

Live demo <https://contextawareadbreaks.centralindia.cloudapp.azure.com>

GitHub https://github.com/ushnardra/HOICHOI_HACKATHON_ADBREAK

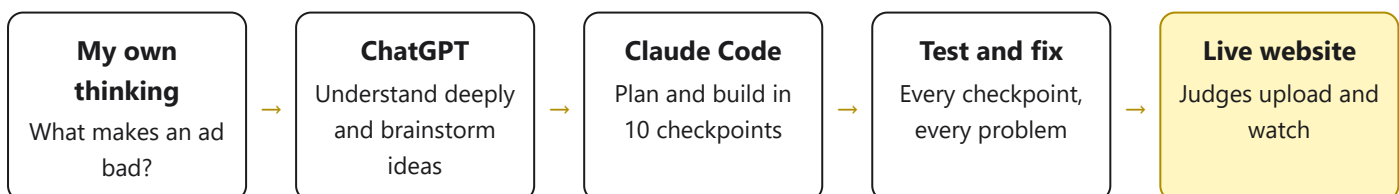
1. The problem in simple words

Streaming apps need ads, but a badly placed ad ruins the story: it cuts someone off mid-sentence, it interrupts a song, or it plays a food ad right after a funeral. The task is to take a long Bengali drama and place ads the way a good human editor would.

Question	What it means
WHERE	Is this a natural, non-jarring place to cut? Cutting in the middle of a sentence is heavily penalised.
WHETHER	Should there be a break here at all? This follows pacing rules: breaks per hour, the gap between breaks, and the total ad time.
WHAT	Which brand fits this moment? The scene's main activity decides it. A brand's negative contexts are a hard block , not just a lower score.

What the judges want: the video goes in, scenes are found, breaks are scored, and brands are matched (a new 9th brand must work with no code change). What comes out is a **VMAP** ad file, a **debug JSON**, and a **demo that really cuts to the ad and comes back**.

2. My journey, from start to end



1 My own thinking first

- I read the problem and thought as a **viewer**: when does an ad feel wrong?
- My list: mid-sentence cuts, cuts inside a song, too many ads, and a wrong ad for the mood (for example, food after a death scene).
- I set my own rules: use **only free tools**, and it must work on **any** episode, not just the sample ones.

2 ChatGPT: understand the problem and brainstorm

- I worked **together with ChatGPT** to understand every part of the problem in depth: what VMAP is, what "negative contexts" means, and what counts as a natural pause.
- We brainstormed ideas:
 - A natural pause = a **scene change** + a **real silence** + a **finished sentence**.
 - To know if a scene is sad or unsafe, use both the **picture** and the **spoken words**.
 - Match brands by **meaning**, but block unsafe brands **completely**.
 - Which free tools and models could do each job.
- I picked the ideas that made the most sense to me and went deeper on them.

3 Claude Code: plan and build

- I used **Claude Code** to turn the ideas into a clear **plan of 10 checkpoints** and to write the code.
- **The rule for every checkpoint:** build it, test it on a real episode, and write a simple report. If something is wrong, fix it and test again. **Only move to the next checkpoint when the current one fully works.**
- This stopped me from building a big app that breaks in many places at once. Every part was checked before the next one used it.

Why the checkpoint method matters: each step is small, tested and proven before the next step builds on it. That is why the final system works on new episodes, with **zero brand-safety mistakes** across 6 episodes.

4 Smart choice with free tools

- The **light AI models** run on my own server: scenes, music, brand matching and speech detection.
- The **heaviest job**, Bengali speech-to-text, uses **Groq's free Whisper large-v3 API**. A full episode is done in under a minute, and it is free to use.

5 Make it live for the judges

- I built a website with two screens: **01 Upload an episode** and **02 Watch with ad breaks**.
- I put it online on an **Azure** server (student account), with a secure **https** link and a proper name: **contextawareadbreaks**.
- It runs all the time, so judges can open it and upload their own video at any time.

6 Final testing

- I tested it on **6 different episodes** (20 to 39 minutes). Every one passed the safety check.
- I added a **9th brand** the system had never seen. It worked with **no code change**.

3. The 10 checkpoints

#	Checkpoint	What it does	Result on the test episode
1	Read the video	Gets the length, frame rate and a clean audio track.	25:53 video, audio ready
2	Camera cuts	Finds every place the camera changes.	228 shots
3	Who is speaking	Finds exactly when people talk, and where the silences are.	Speech and silence mapped
4	Bengali words	Turns speech into words, with the exact time of every word.	Over 1,000 words, under 1 min
5	Scenes	Joins camera shots into scenes using colour, picture and sound.	228 shots → 37 scenes
6	Understand each scene	Labels the activity, place and mood, and flags sad or unsafe scenes (death, illness, violence).	Sensitive scenes flagged
7	WHERE	Only scene changes that are real silences, after a finished sentence, and not inside a song.	0 cuts mid-sentence
8	WHETHER	Pacing: 8 breaks/hour, at least 4 minutes apart, at most 10% ad time, and no break in the first 5 min or last 3 min.	4 breaks chosen
9	WHAT	The best brand for the scene. Unsafe brands are hard-blocked; if no brand is safe, a house promo plays. A final safety check runs on every ad.	Safety check PASS
10	Output + player	The VMAP ad file, the debug JSON, and the website player.	Plays ads and resumes

4. Problems I faced and how I fixed them

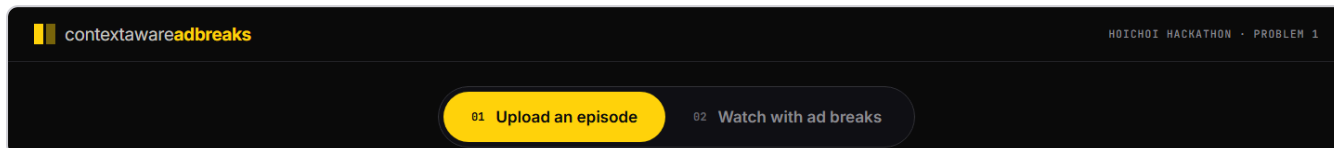
Where	Problem	Fix
Camera cuts	Camera flashes and lightning were counted as cuts.	A flash filter: if the picture jumps and comes back within a few frames, it is not a real cut.
Speed	A full episode took about 15 minutes, too slow for judges.	Moved speech-to-text to Groq's free API. A full episode now takes about 4 min (laptop) or 5–7 min (server).
Word timing	Sometimes one word was stretched across a long silence (once 26 seconds), so silence looked like speech.	Every word is trimmed to the real speech found by the speech detector.
Scene cuts	Cutting at every scene change would have interrupted someone talking up to 73 times per episode.	A scene change is only a candidate. It must also be a real silence after a finished sentence.
Songs	An early version put a break inside a folk song (the song ran across many camera cuts).	A music detector: never cut inside a continuous song, only in a real pause.

Funeral trap	A funeral-ritual meal <i>looks</i> like cooking, so a food ad seemed perfect.	The system also uses the words: "শ্রাদ্ধ" (funeral rite) marks the scene as sensitive, and all food brands are blocked.
Short videos	Short clips got zero breaks.	The no-break zones at the start and end now shrink for short videos.
Hosting	My laptop cannot stay online all the time.	Moved it to an Azure server with a secure https link that runs 24/7.

5. How the website works: a step-by-step UI tour

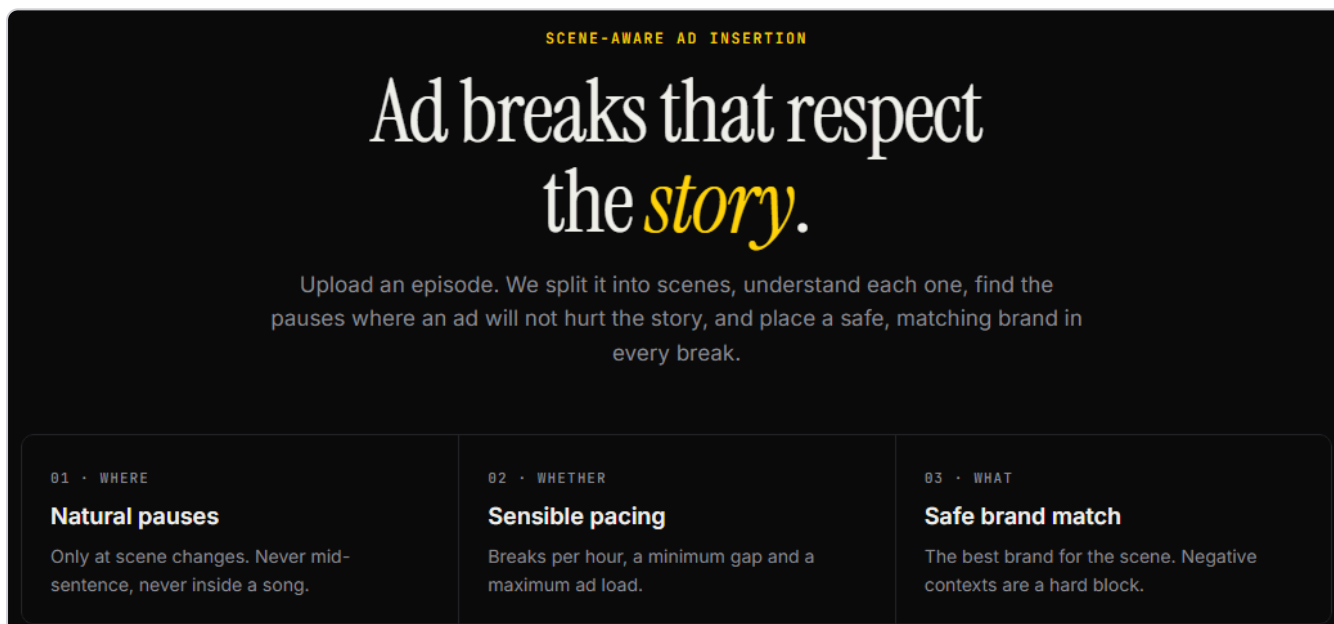
Open <https://contextawareadbreaks.centralindia.cloudapp.azure.com> in any browser. No login is needed.

A The top bar and the main switch



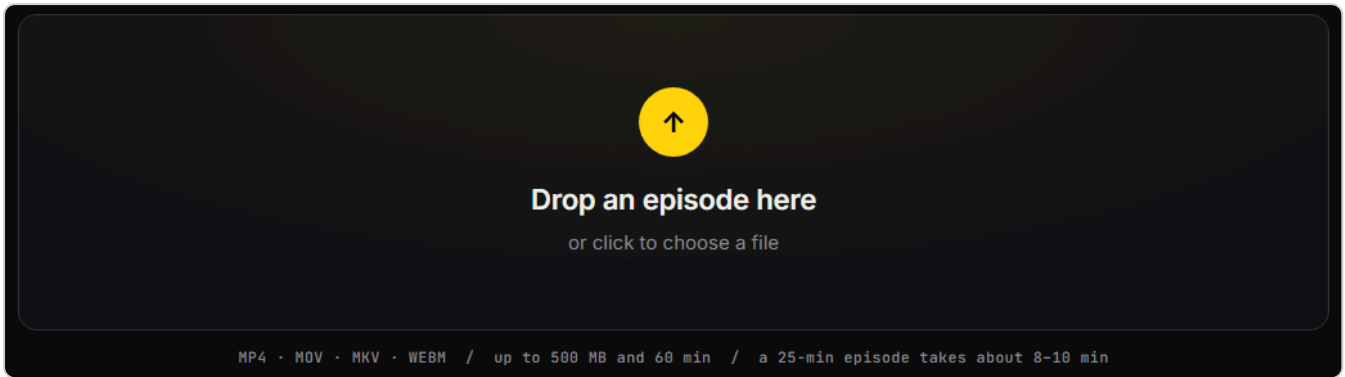
1. **Logo, "contextawareadbreaks"** (top left): click it anytime to go back to the home page.
2. **"Hoichoi Hackathon · Problem 1"** (top right): the event and the problem this project solves.
3. **01 Upload an episode**: the upload screen. It turns yellow when you are on it.
4. **02 Watch with ad breaks**: the player screen, where you watch an episode with its ads.

B The home page: what the product does



1. **Headline**: "Ad breaks that respect the story", the main idea in one line.
2. **Short text**: the whole process in one sentence: split into scenes, understand each scene, find pauses, place a safe brand.
3. **Three cards**, one for each question in the problem: **01 WHERE** (natural pauses, never mid-sentence or inside a song), **02 WHETHER** (pacing rules), **03 WHAT** (safe brand match, with negative contexts as a hard block).

C Upload a video



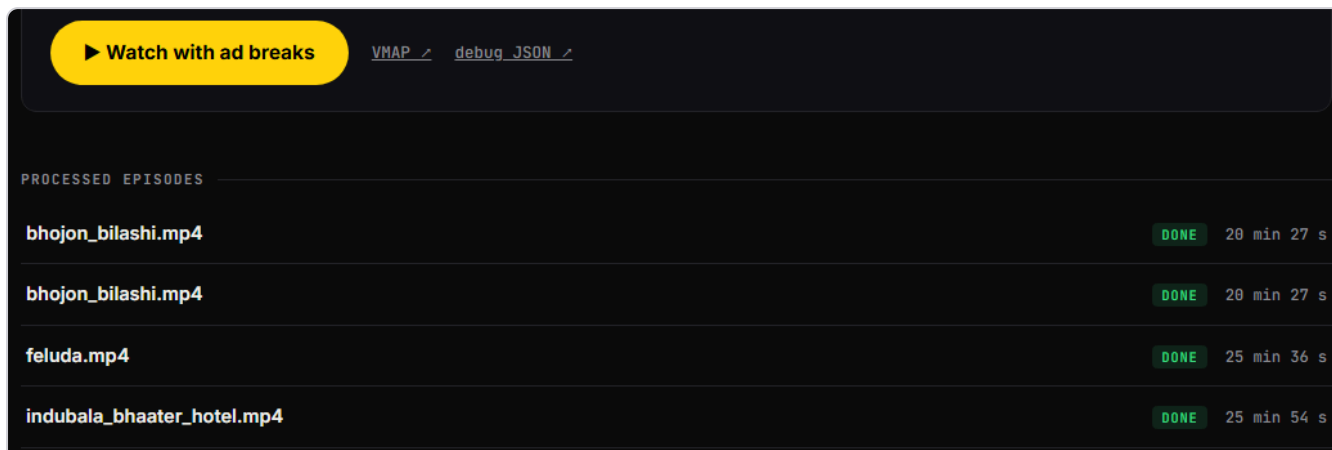
1. **Drop an episode here:** drag a video file onto this box, or click it to choose a file from your computer.
2. **The line below the box** shows the rules: MP4, MOV, MKV or WebM, up to **500 MB** and **60 minutes**. A 25-minute episode takes about **8–10 minutes** in total.
3. After you drop a file, a **yellow progress bar** shows how much has uploaded. The upload speed depends on your internet.
4. If the file is wrong (not a video, no sound, too long), a **red message** explains why.

D Watch the 12 steps run live

Done. Ad breaks are ready			25 min 36 s of video · 6 min 57 s
01	✓	Reading the video and its audio	2 s
02	✓	Finding camera cuts	24 s
03	✓	Finding when people speak	11 s
04	✓	Transcribing Bengali speech	33 s
05	✓	Taking pictures of every shot	22 s
06	✓	Joining shots into scenes	2 min 33 s
07	✓	Finding songs and music	2 min 31 s
08	✓	Understanding each scene (safety)	9 s
09	✓	WHERE: finding natural pauses	1 s
10	✓	WHETHER: choosing breaks (pacing)	0 s
11	✓	WHAT: matching safe brands	11 s
12	✓	Writing the VMAP and debug JSON	0 s

1. **Title:** "Processing <file name>" while it works, then "Done. Ad breaks are ready".
2. **Top right:** the video length (25 min 36 s here) and the time taken so far (6 min 57 s here).
3. **12 numbered steps**, the whole pipeline. The step that is working now is **highlighted in yellow**. A finished step gets a **green tick ✓** and shows how long it took.
4. The steps in order: read the video → camera cuts → who is speaking → Bengali words → pictures of every shot → scenes → songs and music → understand each scene (safety) → **WHERE** → **WHETHER** → **WHAT** → write the VMAP and debug JSON.
5. If more people upload at the same time, you will see **"Waiting in the queue"** first. One video is processed at a time.

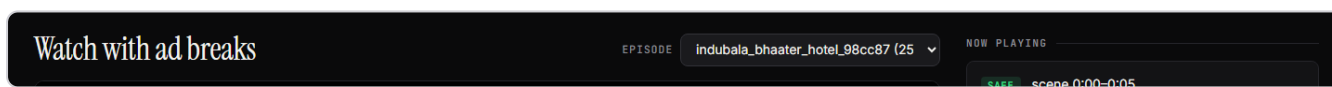
E When it is done



Episode Name	Status	Duration
bhojon_bilashi.mp4	DONE	20 min 27 s
bhojon_bilashi.mp4	DONE	20 min 27 s
feluda.mp4	DONE	25 min 36 s
indubala_bhaater_hotel.mp4	DONE	25 min 54 s

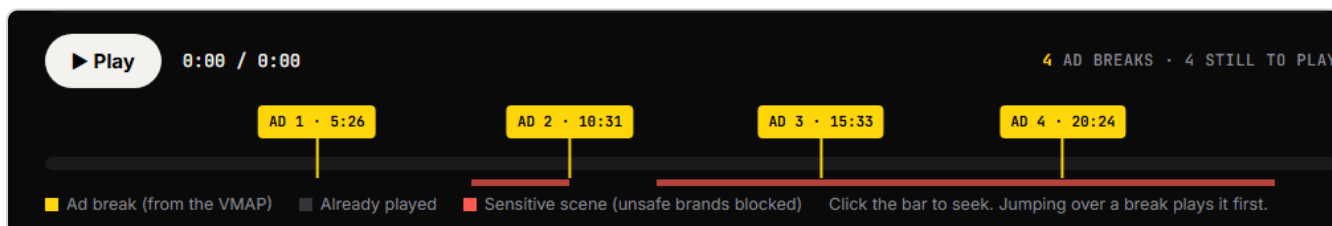
1. ► **Watch with ad breaks** (yellow button): opens this episode in the player.
2. **VMAP** ↗: the standard ad-break file (VMAP 1.0). It lists the time of every break and the ad to play.
3. **debug JSON** ↗: every decision with its reason: every scene, every possible break and why it was used or rejected, and which brands were blocked and why.
4. **Processed episodes**: every finished episode. Click a name to watch it. **DONE** means ready, and the time on the right is the episode length.

F The player: choose an episode



1. **"Watch with ad breaks"**: the player screen title.
2. **EPISODE drop-down** (right): pick any processed episode. The page loads its video and its ad breaks.

G The timeline: where the ads are



1. ► **Play / II Pause**, and the current time out of the total time.
2. **"4 AD BREAKS · 4 STILL TO PLAY"**: how many breaks this episode has, and how many you have not watched yet.
3. **Yellow pins "AD 1 · 5:26"**: each pin is one ad break from the VMAP file, with its time. A pin turns grey after that ad has played. In the last 5 seconds before a break, the pin **pulses**.
4. **Red bars** under the timeline: **sensitive scenes** (death, illness, violence). Unsafe brands are blocked near them.
5. **Click anywhere on the bar** to jump there. If you jump over a break, that ad plays first, and then the video continues where you clicked.

H The side panel: why each ad was chosen

The screenshot shows a video player interface with a dark theme. The side panel is divided into three main sections: 'NOW PLAYING', 'NEXT AD', and 'ALL AD BREAKS'.
1. 'NOW PLAYING': Shows a green 'SAFE' label, the scene time '0:00-0:05', and the activity 'title_card · title_card · tense'.
2. 'NEXT AD': Shows 'Dhoop Deep Incense in 5:26 (at 5:26)' in a yellow box.
3. 'ALL AD BREAKS': Lists four breaks with timestamps in yellow boxes:
- Break 1 (5:26): 'Dhoop Deep Incense'. Description: 'best match for the dominant activity 'work_business' (score 0.346); 0 brand(s) blocked by negative contexts'. Button: 'Watch from 8 s before'.
- Break 2 (10:31): 'House promo (no safe brand)'. Description: 'every brand was blocked by its negative contexts'. Red text: '8 brands blocked by scene safety (hover for names)'. Button: 'Watch from 8 s before'.
- Break 3 (15:33): 'Sanchay Savings Bank'. Description: 'best match for the dominant activity 'prayer_ritual' (score 0.067); 7 brand(s) blocked by negative contexts'. Red text: '7 brands blocked by scene safety (hover for names)'. Button: 'Watch from 8 s before'.
- Break 4 (20:24): 'Sanchay Savings Bank'. Description: 'best match for the dominant activity 'cooking' (score 0.104); 7 brand(s) blocked by negative contexts'.

1. **NOW PLAYING:** the scene on screen right now, with a **SAFE** (green) or **SENSITIVE** (red) label, the scene's start and end time, and its **activity · setting · mood**. For a sensitive scene it also shows the reason, for example "words: death 'মৃত' at 22:57".
2. **NEXT AD:** which brand comes next and in how many seconds.
3. **ALL AD BREAKS:** one card per break:
 - the **time** (yellow box) and the **brand name**,
 - **why** this brand won, for example the scene's main activity and its match score,
 - "**N brands blocked by scene safety**" in red. Hover over it to see which brands were blocked,
 - ► **Watch from 8 s before:** jumps to 8 seconds before the break, so you can see the scene end and the ad begin.
4. "**House promo (no safe brand)**": every brand was blocked here, so the channel's own promo plays instead.

I During playback and during an ad

The screenshot shows a video player interface. The main video area displays a scene with a man in a library, with a 'SMOKING KILLS' warning in the bottom left. A large diagonal watermark 'HOICHOI HACKTATION COPY' is visible. The player controls at the bottom include a play button, a progress bar showing 21:36 / 25:35, and a '4 AD BREAKS · 0 STILL TO PLAY' indicator. The right sidebar contains a 'SENSITIVE' scene notification for the time range 21:32-24:34, listing activities like 'work_business', 'home_room', and 'sad', and words like 'death'. Below this is a 'NEXT AD' section stating 'No more ad breaks'. The 'ALL AD BREAKS' section lists four breaks: Break 1 (Dhoop Deep Incense) at 6:35, Break 2 (Pothik Rail Travels) at 11:24, Break 3 (Cha-Adda Tea) at 15:24, and Break 4 (Sanchay Savings Bank) at 21:33. Each break includes a 'Watch from 8 s before' button and a brief description of the match for the dominant activity 'unclear'.

1. As the video plays, **NOW PLAYING** updates for every scene. Here the scene is **SENSITIVE** because a death word was heard.
2. 5 seconds before a break, a small "**Ad break in 5... 4... 3...**" countdown appears in the corner of the video.
3. At the break, the episode **pauses** and the ad plays on top of it:
 - a yellow **ADVERTISEMENT** tag, the **brand name**, and the **time left**,
 - a "**Why here**" box that explains the choice,
 - a yellow **progress bar** at the bottom that shows the ad's progress,
 - a **Skip ad** button, for the demo only.
4. When the ad ends, the episode **continues from the exact same frame**, so nothing is missed.

6. Negative contexts: the safety rule (example: Indubala Bhaater Hotel)

What is a "negative context"? Every brand in the catalogue lists the moments it must **never** appear next to. If the scene before or after a break touches one of these, that brand is **hard-blocked**. It is not given a lower score; it is removed completely. If every brand is blocked, the channel's **house promo** plays instead. Before anything is saved, a final **safety check** re-checks every ad.

Brand (invented name)	Type	Must NEVER appear next to (negative contexts)
Ranna Rani Masala	Food: spices	death, funeral, mourning, illness, violence, morgue
Bhaat Bhandar Rice	Food: rice	death, funeral, mourning, funeral meal, illness, violence
Cha-Adda Tea	Tea	death, funeral, violence, hospital
Jhalmuri Junction Snacks	Snacks	death, funeral, mourning, sadness, illness, violence, hospital
Pothik Rail Travels	Travel	death, accident, violence, railway accident
Sanchay Savings Bank	Savings	violence, crime
Dhoop Deep Incense	Prayer items	violence, eating food, party
Nirapod Health Cover	Health insurance	death, funeral, violence, morgue

1 The funeral trap: break at 20:24



What the picture shows: a courtyard with a clay cooking stove. The system labels the scene "**cooking**".



What is said (21:59): "শাদা", the Hindu funeral rite. The woman is in a white sari, a sign of mourning.

1. Picture → cooking. A food ad would look like the perfect match.

2. Words → "শাদা" is heard, so the scene is marked **SENSITIVE: death**. The meal is for a funeral.

3. Hard block → **7 brands removed:** all food brands (masala, rice, snacks), tea, travel and health insurance (because of death), and incense (because of "eating food").

4. Result → the only safe brand left, **Sanchay Savings Bank**, plays. **No food ad next to a funeral.**

This is exactly the trap in the problem statement ("a food ad after a funeral"). A system that only looks at the picture would fail here. Mine uses the picture **and** the words.

2 When nothing is safe: break at 10:31



What the picture shows (10:22): railway tracks with things scattered on them, an accident scene. The system marks it **SENSITIVE: violence**.

- The food, tea, health and incense brands are blocked because of "**violence**".
- **Pothik Rail Travels** is blocked because "**railway accident**" is seen in the pictures. A train-travel ad next to a train accident would be terrible.
- **Sanchay Savings Bank** is blocked because of "**crime**" (violence).

Result: all 8 brands are blocked, so the system plays a **House promo** instead of taking any risk.

7. Context-aware brand matching (example: Feluda)

How the best brand is chosen: every brand also lists **target contexts**, the moments it fits best. The system compares the **meaning** of these with what is happening in the scene (activity, place, mood and words) and gives each brand a **match score**. After the unsafe brands are removed, the **highest score wins**. The ads are **never random**, and the same video always gets the same ads.

3 A travel scene gets a travel ad: break at 11:24



10:50: the characters are travelling in a car. The scene is labelled "**travel**".



11:19: the car drives down a country road, just before the break. Labelled "**travel**" and "**street / railway**".

Brand	Target contexts	Match score
Pothik Rail Travels ✓ winner	a train journey, travelling on a road , a busy street, a trip to the countryside	0.28
Cha-Adda Tea	friends or family talking, a relaxed conversation, a calm evening	0.15
Dhoop Deep Incense	a quiet prayer, a peaceful home at night, a religious ritual	0.01

Result: nothing is sensitive here, so no brand is blocked, and the travel brand **Pothik Rail Travels** wins because the scene is a road journey. **A travel ad for a travel scene.**

Same system, different scene, different ad

Break	Scene	Ad chosen	Why
Feluda 11:24	Road journey (travel)	Pothik Rail Travels	Best match: "travelling on a road"
Feluda 15:24	A calm, normal scene	Cha-Adda Tea	Best match: a relaxed, calm moment
Indubala 20:24	Cooking for a funeral (শ্রদ্ধা)	Sanchay Savings Bank	All food brands hard-blocked by "death / funeral"
Indubala 10:31	Railway accident (violence)	House promo	All 8 brands blocked, so no risk is taken

You can check all of this yourself on the live site: open the player, choose the episode, and click **Watch from 8 s before** on the break. The reasons are in the side panel and in the debug JSON.

8. How it follows the rules

Rule	Status
No hard-coded timestamps or brand choices	✓ Everything comes from the video and the config files. Tested on 6 episodes.
No negative-context mistakes	✓ Hard block and a final safety check. 0 mistakes across 6 episodes.
Demo runs live	✓ Online 24/7 on Azure with https.
Synthetic brand names only	✓ All 8 brands (plus the test 9th brand) are invented names.
A 9th, unseen brand works with no code change	✓ Add one entry to the brand file and it works.

9. Tools I used

AI helpers

- **My own thinking**: the problem from a viewer's side, and my rules.
- **ChatGPT**: understanding the problem and brainstorming ideas.
- **Claude Code**: the 10-checkpoint plan and the code.

Free tools in the product

- **ffmpeg**: reads the video and audio.
- **Silero**: finds speech.
- **Groq Whisper large-v3** (free API): Bengali words.
- **SigLIP**: understands the pictures. **AST**: finds music and songs.
- **MiniLM**: matches brands to scenes.
- **FastAPI, Docker, Caddy, Azure**: the website and the server.

All brand names are invented. The pacing numbers are demo settings, not official Hoichoi rules.